
Image Analysis

An Open-Source Summary

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1 Introduction

The key take away is that vision is subjective to the human being and themselves. It is the results of continuous evolution and survival of the fittest. We are more sensitive to light and certain colors, to motion, ... And this fact will be used for efficient image representation and analysis.

We can think of the classic illusion where our brain will interpret certain lengths of an arrow longer than others. Context also does matter, even in a blurry picture our brain will train to make up for what he sees.

1.1 The basics

We first need light to have vision. Light is influenced by object and create perception. ADD image slide 51. The light source bounce on the object and some scattered ray will get trough the 2D Image plane, which will be casted on the sensor array thanks to the lens system.

1.1.1 Light

We will look at it from a physical optic (not geometrical (constant speed and straight path) nor quantum). Light is an Electro-magnetic wave which we assume it to be planar. aka $\frac{|E|}{|H|} = \eta$ in simple term, light is perpendicular. This is a **self-sustaining process** that is characterized by:

1. λ : wavelength
2. Direction
3. E : Amplitude
4. φ : phase
5. Direction of polarisation

Violet: low frequency ($380nm$); Red: high frequency ($760nm$). Remember $\lambda = \frac{c}{f}$.

1.1.2 Perception

We don't perceive light equally. Humans have 3 cones that are not evenly spread apart. But the human vision is trained to compensate for it so everything feels similar in intensity to us. Some animals can have more cones.

1.1.3 Interaction with matter

Table 1: The four types of interaction

Phenomenon	Example
Absorption	Blue water, green leaf
Scattering	blue sky, red sunset
Reflection	colored in k
Refraction	dispersion by a prism

- Absorption: Dissipation of λ specific for the medium. Resonance in molecules.
- Scattering: When light (radiation/particles) deviates from a straight trajectory by local non-uniformities. Type depends on the relative size of particles and

We ignored transmission and diffraction.

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2.1 Modification

To contribute to this work or any other one from this project please find more information at the Github repository.

2.2 Take down

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